

Question:

Kylar went to the store to buy glasses for his new apartment. One glass costs \$5, but every second glass costs only 60% of the price. Kylar wants to buy 16 glasses. How much does he need to pay for them?

Reasoning + Answer:

Let's break down the problem step by step to determine how much Kylar needs to pay for 16 glasses .

1 . ** Understanding the pricing structure : **

- The cost of one glass is \$ 5 .
- The cost of every second glass is 60 % of the price of one glass .

2 . ** Calculating the price of every second glass : **

- To find 60 % of \$ 5 , we use the following calculation :

$$\begin{aligned} & \backslash[\\ & 0.60 \times 5 \\ & \backslash] \end{aligned}$$

- Performing the multiplication :

$$\begin{aligned} & \backslash[\\ & 0.60 \times 5 = 3 \\ & \backslash] \end{aligned}$$

- Therefore , the cost of every second glass is \$ 3 .

3 . ** Determining the pattern of glasses : **

- Kylar wants to buy 16 glasses .
- The pattern alternates between \$ 5 and \$ 3 .
- We can pair the glasses to simplify the calculation :
- The first glass costs \$ 5 .
- The second glass costs \$ 3 .
- The third glass costs \$ 5 .
- The fourth glass costs \$ 3 .
- And so on...

4 . ** Counting the number of pairs : **

- Since Kylar is buying 16 glasses , we can divide the total number of glasses by 2 to find the number of pairs :

$$\begin{aligned} & \backslash[\\ & \frac{16}{2} = 8 \\ & \backslash] \end{aligned}$$

- This means there are 8 pairs of glasses .

5 . ** Calculating the total cost for the pairs : **

- Each pair consists of one glass at \$ 5 and one glass at \$ 3 .
- The total cost for one pair is :

$$\begin{aligned} & \backslash[\\ & 5 + 3 = 8 \\ & \backslash] \end{aligned}$$

- Since there are 8 pairs , we multiply the cost of one pair by 8 :

$$\begin{aligned} & \backslash[\\ & 8 \times 8 = 64 \\ & \backslash] \end{aligned}$$

6 . ** Final calculation : **

- The total cost for all 16 glasses is :

$$\begin{aligned} & \backslash[\\ & 64 \\ & \backslash] \end{aligned}$$

Thus , the amount Kylar needs to pay for 16 glasses is :

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1 . Cost of one glass : \$ 5

2 . Cost of every second glass : $0.60 \times \$ 5 = \$ 3$

3 . Pattern : \$ 5 , \$ 3 , \$ 5 , \$ 3 , ...

4 . Number of pairs : $16 / 2 = 8$

5 . Cost per pair : $\$ 5 + \$ 3 = \$ 8$

6 . Total cost : $8 \times \$ 8 = \$ 64$

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<answer> 64 </answer>